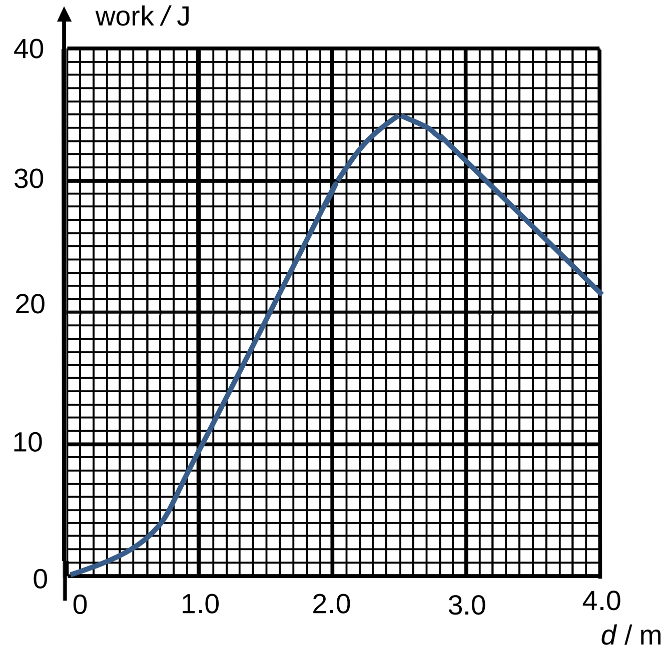
<table>
<tr>
<td>2(a)</td>
<td>Work done is area under the F-d graph.

<table>
<tr><th>d / m</th><th>work done / J</th></tr>
<tr><td>1.0</td><td>10</td></tr>
<tr><td>2.0</td><td>30</td></tr>
<tr><td>2.5</td><td>35</td></tr>
<tr><td>4.0</td><td>35 – 13.5 = 21.5</td></tr>
</table>

Note that work done is cumulative.</td>
<td>[1]
[1]

[1]
[1]

One mark each for the correct shape and value corresponding to each d value.</td>
</tr>
</table>

	 **Fig. 2.2**	
2(b)(i)	$\sin 30° = \frac{h}{d}$ $mgh = mgd \sin 30° = (0.50)(9.81)(2.0)\sin 30° = 4.9$ J	[1] answer
2(b)(ii)	Gain in K.E. = W.D. on mass – W.D. against friction – gain in G.P.E. Gain in K.E. = 30 – (3.0)(2.0) – 4.9 = 19.1 J	[1] correct energy equation [1] correct W.D. again friction [1] correct answer
3(a)	Acceleration is a rate of change of velocity	[1]